K. J. Somaiya Institute of Technology

Department of Artificial Intelligence and Data Science

Project-Based Learning (PBL) – One Page Project Summary

Academic Year: 2026-27 | Semester: V

1. PROJECT DATA

Project Title: Indian Meteorological Department Aviation Forecast Verification Web Portal

Project Domain: Data Science and Web development

Project Guide: Pravin Patil

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2. EXECUTIVE SUMMARY

The Aviation Forecast & Verification Web Portal is a web-based application developed during an internship at the India Meteorological Department (IMD) to automate the verification of aviation weather forecasts. The system compares Take-Off Forecasts, Local Area Forecasts, and Aerodrome Warnings with observed METAR and upper-air data to evaluate forecast accuracy. It generates automated reports, accuracy metrics, and graphical visualizations, significantly reducing manual effort while improving the efficiency and reliability of aviation forecast verification.

3. PROBLEM STATEMENT

The existing aviation forecast verification process at the India Meteorological Department (IMD) is largely manual, making it time-consuming, repetitive, and prone to human errors. There is a need for an automated system that accurately verifies Take-Off Forecasts, Local Area Forecasts, and Aerodrome Warnings against observed weather data while generating performance reports for efficient forecast evaluation.

4. KEY OBJECTIVES

- To automate the verification of aviation weather forecasts using METAR and upper-air observations.
- To evaluate the accuracy of Take-Off Forecasts, Local Area Forecasts, and Aerodrome Warnings based on standard IMD verification criteria.
- To generate parameter-wise accuracy reports, graphical analyses, and performance metrics for forecast evaluation.
- To reduce manual effort and support faster, more reliable, and consistent aviation forecast verification.

5. METHODOLOGY & TECH STACK

Algorithms: Rule-based forecast verification, data preprocessing, data extraction, data cleaning, time-based data matching, parameter-wise comparative analysis, and accuracy computation using IMD verification criteria.

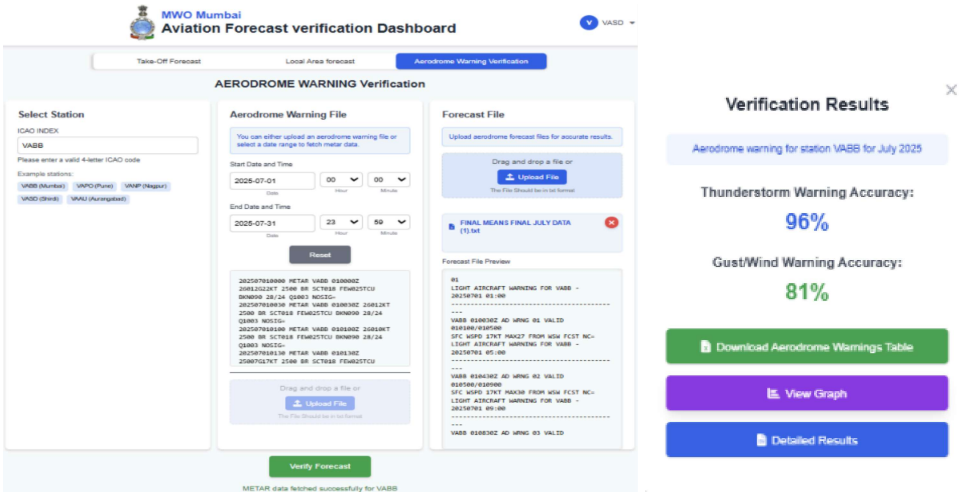
Tools & Frameworks: Python, Flask, Pandas, HTML, CSS, JavaScript, Bootstrap, SQLite, Docker, Nginx, Git, GitHub.

Dataset Used: IMD aviation forecast data (Take-Off Forecasts, Local Area Forecasts, Aerodrome Warnings), METAR observations, and Upper-Air observation data.

Hardware/Environment: Windows, Visual Studio Code, Docker containers, Nginx, SQLite, and deployment on the IMD server.

6. OUTCOMES & DELIVERABLES

Working Prototype/Model: A fully functional web-based Aviation Forecast Verification System developed and deployed on the IMD server. The system automates the verification of Take-Off Forecasts, Local Area Forecasts, and Aerodrome Warnings using METAR and Upper-Air observations, and generates accuracy reports, tables, and graphical visualizations.



Code Repository: https://github.com/aryanilake/forecast_dashboard.git

Key Findings:

- Successfully automated the aviation forecast verification process, reducing manual effort and verification time.
- Improved the consistency and accuracy of forecast evaluation using standard IMD verification criteria.
- Generated parameter-wise and overall forecast accuracy reports with graphical analysis.
- Enabled faster identification of forecast errors, supporting data-driven improvements in aviation weather forecasting.

Project Guide Name and sign: